2021 International Conference on Computing, Communication and Green Engineering (CCGE)
JSPM's RSCOE, Pune, India. Sep 23-25, 2021

Security of User Credentials on Web Portals

Amit Hire
Pimpri Chinchwad College of Engineering,
Pune, India
amit.hire@pccoepune.org

Jayashree Katti
Pimpri Chinchwad College of Engineering,
Pune, India
jayashree.katti@pccoepune.org

***Abstract*— The most prevalent form of authentication or computer system intrusion protection is the use of alphanumeric usernames and passwords. Strong password schemes have traditionally provided a certain level of protection, but they are complicated and difficult to remember, prompting their owners to write them down on paper. As a consequence, protection is jeopardized.Increasing the sharing of information, internet use, data transfer, protection and validation of passwords have become significant challenges.As a result, many programmes use different hashing and salting algorithms to store password files in databases.Attackers, on the other hand, do not consider password hashing to be safe since they use brute force attacks, password guessing attacks, and other methods to obtain the user's password from the database's password log.Some systems store passwords in databases using honeywords or bogus passwords generated using the generation technique for honeywords to prevent hacker attacks.In our recommended approach, passwords are saved using a distinct hashing mechanism with a less complexity of time in which majority of the processes need basic operation based on binarys.**

Keywords-Web Security;User Credential; Hash Algorithm, Database Security;Authentication.

I. INTRODUCTION

The first layer of encryption is often employed with password-based authentication, which means every assailant must guess the password line at the fewest possible turns. As security professionals build systems for password authorisation that prohibit hackers from splitting the password. Using case-sensitive characters including one or many random characters in a password string might raise the complexity of the password. In limited iterations, hackers utilise advanced technologies for cracking a password.. As a result, the complexity of the string is not adequate to safeguard the system.

As a end result, the password authentication machine has to be stepped forward.To shield the privateness of a user's credentials and to save you any unauthorized usage due to fraud or leakage, it's essential to cozy consumer comments. The most extensively used kind of authentication and the most vulnerable to a number of forms of attacks is password-based authentication.There have recently been significant security breaches to rob user credentials, leading to huge economic losses to consumers and financial institutions as large numbers of credentials have been stolen..Data breach damage may be enormous. Besides the obvious reputational harm, remediation and penalties expenses and sales losses, indirect costs may be enormous.

On average, employee email credentials in some sort of data breach were found to have been exposed from 10 percent of Fortune 500 businesses. An alarming 23% of employee emails have been discovered in data leaks in the telecoms industry while the energy and finance industries report 18% and 17% leaks, respectively.

In addition to the company's risk for shame and extortion owing to various acts of indiscretion, leaked credentials can and will be available online because of the more common practise of improper password management.

In many cases employees will use one password variant on a number of internet accounts, ranging from 'Johndoe1' to JohnDoe2' and 'JohnDoe123,' etc. A number of passwords may be used in email, banking, online shopping and other activities that are not so distinct.

The failure of a set of credentials could provide cyber thieves with sufficient information for the purposes of rebuilding a whole digital identity.

Passwords should not be too short, nor should they include words from the dictionary, and they should be updated regularly.

Password Definition

To create a strong password, use the following rules :

"A decent password should contain the following: -

1. Both case letters i.e lower and upper case, letters and/or digit and/or punctuation characters
2. Is simple to remember since it must not be recorded,
3. Words that aren't included in a dictionary.
4. It can't be a friend's name, a relative's movie star name, or even a character from a novel.

The current increase in active ransomware, phishing schemes, and other assaults has demonstrated that having a user name and a static password is inefficient and necessitates more consideration. Authentication techniques are defined by what you know, what you have, and who you are. Single-factor authentication is not seen to be highly secure.

A. Username And Password Authentication

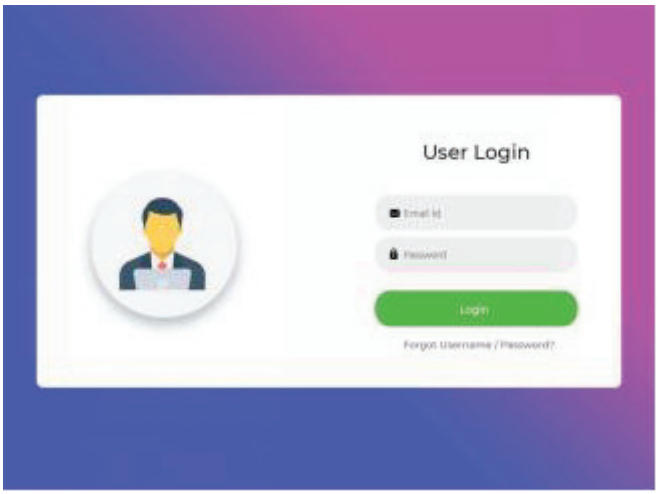
The process of verifying that a user is who they assert to be is known as authentication.

1. What you know - A password, PIN, or personal details such as your birth place, for example.
2. What you have - A tangible object such as a mobile phone or a credit card
3. What you are - Biometric information such as a fingerprint or a retina scan, among other things.

2021 International Conference on Computing, Communication and Green Engineering (CCGE) | 978-1-6654-1509-5/21/$31.00 ©2021 IEEE | DOI: 10.1109/CCGE50943.2021.9776476

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on October 13,2026 at 09:22:30 UTC from IEEE Xplore. Restrictions apply.

The most prevalent type of authentication is password authentication, which has been categorised as what you know.



II. Literature Survey

Kelley et al. [1] investigated the creation of metrics to characterise password intensity and their application to the evaluation of password-composition policies.In an online analysis, 11K+ passwords obtained by the composition of seven passwords were investigated[10]. They also developed an effective distributed approach to determine how many heuristic algorithms can formulate passwords. They looked at passwords that are formed under various circumstances, the relation between passwords that are actually developed according to a given structure policy and the relation between guessability as determined by algorithm for cracking passwords, and estimations of entropy.Their findings could help us better understand password composition policies and password protection solutions.

Komanduri and others [2] evaluated password entropy to determine its predictability and found many commonly held convictions to be wrong.

Agrawal and others. [3], Multidimensional password has been presented as a powerful password generating approach taking into consideration many input cloud paradigm characteristics.

Mohammed [4] proposed a genetic algorithm-based password generator that met all of the specifications of a standard password.

Two Factor Authentication has proven to be resistant to potential attacks on user access privileges misuse. 20+ seconds is sufficient to avoid hackers from logging into the system and controlling against his consent, according to Henki Seta et. al [5] study.

With advancements in graphical processing unit (GPU) technology, Ziya et al. addressed how passwords are becoming easier to crack. A brute attack on the hash password will allow an assailant to retrieve a password for a user. After the password has been recovered, no server will notice any user authentication incorrect [6].

Many academics have focused much on password security. The standard password cracking strategies are based on models that use password data. Our work provides a new online PWD authentication method by inserting separators in passwords to reinforce the current user authentication system, to reduce this hazard. [7].

A new database of passwords named PassDB which implements a high precision password management system using 3DBF[8]. PassDB follows three levels of architecture, i.e., user room, network room and server room. In the user space, a user uses Murmur hazing to code the User ID, password and password. These 3 values are encrypted and sent over the network using the public key of the server. Three messages which are encrypted are delivered to the server space, which is decrypted using the secret key of the server.

III. Proposed System

In this paper, proposed methodology to provide more secure passwords by extending the hashing algorithm and combining salt values and cryptographic concepts, as well as to make passwords more secure by disorganizing or rearranging the information in an unreadable manner.

Objectives of this paper are:

1. Passwords must be encrypted using appropriate encryption techniques.
2. To reduce the number of people who steal passwords.
3. To reassure users about the security of their passwords.

An existing authentication system is used in the proposed system and added new methods such as an enhanced hashing algorithm and the combination of Salt Values and concept of cryptography. Results provided by the existing system are not very accurate and also consumes more time.

The proposed Scheme Modules consist of -

1. User Registration
2. Adding Salt Values
3. Hash Function

A. User Registration

To register, you must first create a new user account, which is a database record that describes how you can prove your identity. To authenticate you, you must have proof that your identity matches that of your user account.In this user is asked for few details in order to register with its username and password.

B. Encryption/Decryption

Here the passwords are encrypted into unreadable text in order to prevent hackers. Use of cipher text mechanism is done to generate cipher text for further process.

C. Adding Salt Values

Appending some salt values, also referred to as irrelevant data, adds an extra layer of security protection. This lengthens the password, resulting in an additional layer of protection

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on October 13,2026 at 09:22:30 UTC from IEEE Xplore. Restrictions apply.

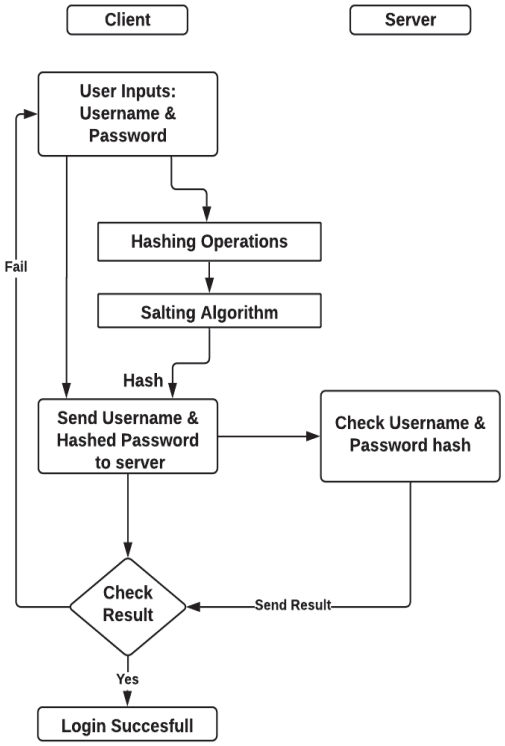


Fig. 1. System Flow Chart

D. Hash Function

To ensure the integrity of transmitted or stored data, a cryptographic hash function is used. It is often referred to as a message digest. The hash function produces a fixed-size message digest of a given message, which is used as the message's signature. MD = HF (M) is a hash function mathematical definition, where HF is a hash function with the following properties.

(a) Since it is a one-way equation, it is simple to calculate MD from M but difficult to calculate M from MD.

(b) Finding messages M1 and M2 that produce the same message digest, HF (M1) 6 = HF (M2), should be difficult (M2).

(c) MD-5 creates a message digest with a fixed length of 128 bits. It takes 64 rounds to complete.

IV. ALGORITHM

The hashing and salting technique in our proposed system is thought to provide faster time for key or password security. The following is the algorithm:

Require: Inputted string which is considered as password.

1. Begin Procedure.
2. Data:Inputted string
3. N:Length of password is calculated
4. R:String is reversed
5. BS:Convert the reversed string into a binary string.
6. PS:Padding bits are calculated depending upon the length of the password.
7. Fill in the gaps in this binary string with padding bits i.e. concat(PB+BS)
8. HS:Replace Padded binary (PS) string to its hexadecimal equivalent.
9. Hashing operation is done on a hexadecimal equivalent password.
10. Hash value generated and saved in database

V. RESULTS AND DISCUSSIONS

TABLE I. RESULTS

Password String	Hash Value
mytest	59a48b3bcf3e1337e1a6ce4509df6b6b
TestPass@123#	507201d9695dee7b157da44fd6dcb02e

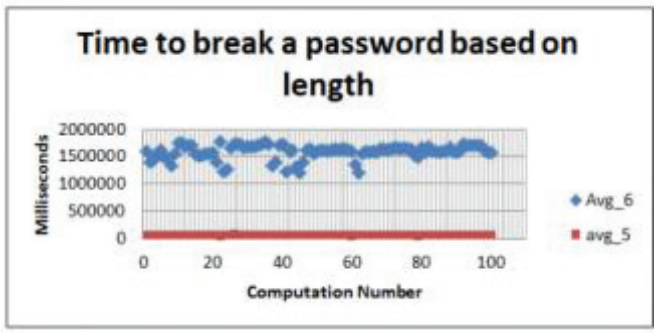


Fig. 2. Time taken to break 5 and 6 letter passwords

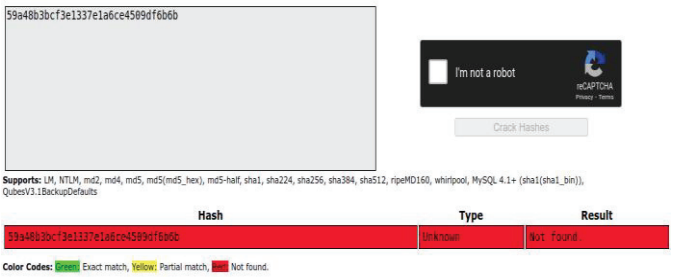


Fig. 3. Hash Value Not Found

TABLE II. COMPARISON TABLE WITH TRADITIONAL HASHING OPERATION.

No	Parameter Comparison	Before	After
1	Login feature security standard	It is unsatisfactory because it continues to use the old hash method, which has been shown to have a dangerous vulnerability	Already accomplished. The encryption update employs hash functions that provide a more reliable and robust level of security.
2	The password security level on the web-based application login mechanism.	Not that good, because the algorithm method employed has proved hazardous	Good, because the method for algorithms was safer and more reliable

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on October 13,2026 at 09:22:30 UTC from IEEE Xplore. Restrictions apply.

3	The total value provided by the hash function.	Since the resulting hash value is non salted and simple, a brute force test only takes a short time	Since the resulting hash value is complicated, a brute force test takes more time.

VI. CONCLUSION

The secure use of user credentials is a big problem for information safety professionals as a result of the high number of frequent assaults, for example the loss of private information, financial losses, anxiety and mistrust of security measures.A system is needed which can safeguard your password and also provide a complete authentication method.

REFERENCES

[1] Kelley, Patrick & Komanduri, Saranga & Mazurek, Michelle & Shay, Richard & Vidas, Timothy & Bauer, Lujo & Christin, Nicolas & Cranor, Lorrie & Lopez, Julio. (2012). Guess Again (and Again and Again): Measuring Password Strength by Simulating Password-Cracking Algorithms. Proceedings - IEEE Symposium on Security and Privacy. 523-537. 10.1109/SP.2012.38.

[2] Saranga Komanduri, Richard Shay, Patrick Gage Kelley, Michelle L Mazurek, Lujo Bauer, Nicolas Christin, Lorrie Faith Cranor, and Serge Egelman. O f passwords and people: measuring the effect of password- composition policies. In Proceedings o f the SIGCHI Conference on Human Factors in Computing Systems, pages 2595-2604. ACM, 2011.

[3] HA Dinesha and VK Agrawal. Multi-dimensional password generation technique for accessing cloud services. Special Issue on: '' Cloud Computing and Web Services", International Journal on Cloud Computing: Services and Architecture (IJCCSA), 2(3):31—39, 2012.

[4] Sura Jasim Mohammed. A new algorithm of automatic complex password generators employing genetic algorithm. Journal o f University o f Babylon, 26(2):295-302, 2018.

[5] H. Seta, T. Wati and I. C. Kusuma, "Implement Time Based One Time Password and Secure Hash Algorithm 1 for Security of Website Login Authentication," 2019 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS), 2019, pp. 115-120, doi: 10.1109/ICIMCIS48181.2019.8985196.

[6] Toshiyuki Masui. Episodas: Das-based password generation using episodic memories. In Proceedings o f the 2018 International Conference on Advanced Visual Interfaces, page 73. ACM, 2018.

[7] Zheng, Wantong & Jia, Chunfu. (2017). CombinedPWD: A New Password Authentication Mechanism Using Separators Between Keystrokes. 557-560. 10.1109/CIS.2017.00129.

[8] R. Patgiri, S. Nayak and S. K. Borgohain, "PassDB: A Password Database Using 3D Bloom Filter," 2019 IEEE 21st International Conference on High Performance Computing and Communications; IEEE 17th International Conference on Smart City; IEEE 5th International Conference on Data Science and Systems (HPCC/SmartCity/DSS), 2019, pp.

[9] Shay, Richard & Komanduri, Saranga & Kelley, Patrick & Leon, Pedro & Mazurek, Michelle & Bauer, Lujo & Christin, Nicolas & Cranor, Lorrie. (2010). Encountering Stronger Password Requirements: User Attitudes and Behaviors. ACM International Conference Proceeding Series. 10.1145/1837110.1837113

[10] M. Weir, Sudhir A, et al 2010. Testing metrics for password creation policies by attacking large sets of revealed passwords. In Proceedings of the 17th ACM conference on Computer and communications security (CCS '10). Association for Computing Machinery, New York, NY, USA, 162–175. DOI:https://doi.org/10.1145/1866307.1866327

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on October 13,2026 at 09:22:30 UTC from IEEE Xplore. Restrictions apply.